

Formulario del examen de mitad de curso

Deep Learning para Business Analytics · Comillas ICADE · Días 01-09

NEURONA Y RED

Neurona

$$y = \sigma\left(\sum_{i=1}^d w_i x_i + b\right) = \sigma(w^\top x + b)$$

Perceptrón multicapa

$$f_\theta(x) = h_L \circ \dots \circ h_1(x), \quad h_\ell(z) = \sigma_\ell(W_\ell z + b_\ell)$$

Sigmoide

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

ReLU

$$\text{ReLU}(z) = \max(0, z)$$

NORMALIZACIÓN

BatchNorm

$$\mu_B = \frac{1}{m} \sum_{i=1}^m a_i, \quad \sigma_B^2 = \frac{1}{m} \sum_{i=1}^m (a_i - \mu_B)^2$$

BatchNorm: salida

$$\hat{a} = \frac{a - \mu_B}{\sigma_B}, \quad y = \gamma \hat{a} + \beta$$

CONVOLUCIÓN

Convolución 2D

$$Y_{i,j,o} = \sum_{u=0}^{k-1} \sum_{v=0}^{k-1} \sum_{c=1}^{C_{in}} X_{i+u, j+v, c} K_{u,v,c,o} + b_o$$

Dimensión espacial de salida

$$H' = \left\lfloor \frac{H + 2p - k}{s} \right\rfloor + 1$$

Max-pooling

$$Y_{i,j,c} = \max_{0 \leq u, v < k_p} X_{i \cdot s + u, j \cdot s + v, c}$$

Average pooling

$$Y_{i,j,c} = \frac{1}{k_p^2} \sum_{u=0}^{k_p-1} \sum_{v=0}^{k_p-1} X_{i \cdot s + u, j \cdot s + v, c}$$

MODELOS SECUENCIALES

RNN

$$h_t = \sigma(W_h h_{t-1} + W_x x_t + b), \quad y_t = w_y h_t$$

ATENCIÓN

Atención escalada

$$\alpha_j = \text{softmax}\left(\frac{q^\top k_j}{\sqrt{d_k}}\right), \quad \text{Att} = \sum_j \alpha_j v_j$$

EMBEDDINGS

Similitud coseno

$$\cos(u, v) = \frac{u^\top v}{\|u\|_2 \|v\|_2} = \frac{\sum_i u_i v_i}{\sqrt{\sum_i u_i^2} \sqrt{\sum_i v_i^2}}$$

Aritmética semántica

$$\phi(\text{rey}) - \phi(\text{hombre}) + \phi(\text{mujer}) \approx \phi(\text{reina})$$

TRANSFORMER

Dimensión por cabeza

$$d_k = \frac{d_{\text{model}}}{H}$$